

Nombre: Pérez Takamaya, Luis Emilio

NL: 26

NT: 16

Fecha: 06/mayo/2026

Sea el conjunto $A = \{1, 2, 3, 4\}$, coloca $\checkmark$ si cumple o $\times$ si no cumple. En caso de que no cumpla justificar mediante símbolos matemáticos.

$$R_1 = \{(1, 2), (2, 1), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

$$R_2 = \{(1, 1), (1, 3), (2, 2), (2, 4), (3, 1), (4, 2), (4, 4)\}$$

$$R_3 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4), (1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R_4 = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (3, 4), (1, 4), (3, 1), (4, 3), (4, 1)\}$$

$$R_5 = \{(1, 2), (1, 3), (2, 4), (4, 3)\}$$

$$R_6 = \{(1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\}$$

$$R_7 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$$

$$R_8 = \{(1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (1, 4)\}$$

$$R_9 = \{(1, 1), (2, 2), (3, 3), (4, 4), (2, 3), (1, 4)\}$$

$$R_{10} = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$$

	R	IR	S	AS	AT	T	IT	Equivalencia	Orden
R_1	$\times$ $\nexists 1R1$	$\checkmark$	$\times$ $\exists 1R3 \wedge 1R4$	$\times$ $\exists 1R1 \wedge 1R2$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 1R2 \wedge 1R3$	No cumple reflexiva	No cumple reflexiva
R_2	$\times$ $\nexists 1R3$	$\times$ $\exists 1R1$	$\times$ $\checkmark$	$\times$ $\exists 1R1$	$\times$ $\exists 1R3 \wedge 1R4$	$\times$ $\exists 1R3 \wedge 1R4$	$\times$ $\exists 1R3 \wedge 1R4$	No cumple reflexiva	No cumple reflexiva
R_3	$\checkmark$	$\times$ $\exists 1R1$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 1R1$	$\checkmark$	$\checkmark$	$\times$ $\exists 1R2 \wedge 1R3$	No cumple simétrica	$\checkmark$
R_4	$\checkmark$	$\times$ $\exists 1R1$	$\checkmark$	$\times$ $\exists 1R1$	$\times$ $\exists 1R2 \wedge 1R3$	$\checkmark$	$\times$ $\exists 1R1$	$\checkmark$	No cumple antisimétrica
R_5	$\times$ $\nexists 1R1$	$\checkmark$	$\times$ $\exists 1R2 \wedge 1R3$	$\checkmark$	$\checkmark$	$\times$ $\exists 1R2 \wedge 1R3$	$\checkmark$	No cumple reflexiva	No cumple reflexiva
R_6	$\times$ $\nexists 1R1$	$\times$ $\exists 2R2$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 2R2$	$\checkmark$	$\checkmark$	$\times$ $\exists 2R2$	No cumple reflexiva	No cumple reflexiva
R_7	$\times$ $\nexists 1R1$	$\checkmark$	$\times$ $\exists 2R1 \wedge 1R2$	$\checkmark$	$\checkmark$	$\checkmark$	$\times$ $\exists 2R1 \wedge 1R2$	No cumple reflexiva	No cumple reflexiva
R_8	$\times$ $\exists 4R4$	$\times$ $\exists 1R1$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 1R1$	$\checkmark$	$\checkmark$	$\times$ $\exists 1R1$	No cumple reflexiva	No cumple reflexiva
R_9	$\checkmark$	$\times$ $\exists 1R1$	$\times$ $\exists 2R3 \wedge 1R2$	$\times$ $\exists 1R1$	$\checkmark$	$\checkmark$	$\times$ $\exists 1R1$	No cumple simétrica	$\checkmark$
R_{10}	$\checkmark$	$\times$ $\exists 1R1$	$\times$ $\exists 1R2 \wedge 1R3$	$\times$ $\exists 1R1$	$\checkmark$	$\checkmark$	$\times$ $\exists 1R1$	No cumple simétrica	$\checkmark$